

Production Incident Postmortem & Incident Handling Dossier

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Document Reference	INC-20260930-CRIT-01
Incident Severity	P1 (Critical - Core Workflow Degradation & Payment Failure)
Environment	Production (api.zecpath.com)
Date & Timestamp	2026-09-30T14:15:00Z to 2026-09-30T14:48:00Z
Auditor / On-Call	DevOps & Reliability Engineering Team
Deliverable File	docs/day_65/INCIDENT_HANDLING_REPORT.md

Deliverable 1: Comprehensive Incident Report

1.1 Executive Summary

At 14:15 UTC, the Zecpath production platform experienced compounding operational disruptions impacting three mission-critical workflows:

API Crash (500 / 502 Bad Gateway): WSGI Unicorn worker timeouts caused by synchronous thread locking during high-volume document ingestion.

SaaS Payment Failures: Stripe and Razorpay webhook processing failures caused subscription state updates to halt, leaving charged customers without access entitlements.

Voice AI Pipeline Outage: Streaming WebRTC media sockets failed during third-party speech-to-text (STT) socket exhaustion, crashing live candidate interview calls.

Total downtime and partial service degradation lasted 33 minutes. Service was restored to 100% availability at 14:48 UTC after deploying hotfixes, isolating worker pools, and executing rollback procedures.

1.2 Log Analysis Breakdown

A. Application Logs (/var/log/zecpath/app.log)

```
[2026-09-30 14:18:22,104] ERROR [core.services.billing:84] Webhook deserialization failed: KeyError: 'customer_email'
[2026-09-30 14:19:01,891] WARNING [core.views:122] Dropping slow audio stream chunk: connection reset by peer
[2026-09-30 14:20:15,310] CRITICAL [core.utils.ai_voice:204] Voice socket pipe broken. Telephony media dropped.
```

Findings: The payment service attempted to read missing metadata fields from updated gateway JSON payloads, causing silent transaction termination before database state updates.

B. Error Logs (/var/log/unicorn/error.log)

```
[2026-09-30 14:16:42 +0000] [18921] [CRITICAL] WORKER TIMEOUT (pid:19042)
[2026-09-30 14:16:42 +0000] [19042] [INFO] Worker exiting (pid: 19042)
[2026-09-30 14:16:43 +0000] [19430] [INFO] Booting worker with pid: 19430
[2026-09-30 14:17:10 +0000] [18921] [CRITICAL] WORKER TIMEOUT (pid:19430)
```

Findings: Unicorn workers exceeded the 30-second execution window. Synchronous file parsing blocked the WSGI event loop, triggering sequential worker termination and replacement.

C. Server / Ingress Logs (/var/log/nginx/error.log)

```
2026/09/30 14:17:15 [error] 1102#1102: *4032 connect() to unix:/run/unicorn.sock failed (11: Resource temporarily unavailable) while connecting to upstream, client: 198.51.100.44, server: api.zecpath.com, request: "POST /api/candidate/resume/parse-structured/ HTTP/1.1", upstream: "http://unix:/run/unicorn.sock:/api/candidate/resume/parse-structured/"
```

2026/09/30 14:17:18 [error] 1102#1102: *4035 recv() failed (104: Connection reset by peer) while reading response header from upstream

Findings: Nginx saturated its connection backlog to the Unix domain socket due to Gunicorn worker timeouts, returning HTTP 502 Bad Gateway responses to incoming API clients.

1.3 Incident Timeline & Triage Log

Timestamp (UTC)	Phase	Event & Action Taken
14:15	Alert	Automated Prometheus alert triggered: Nginx 5xx Error Rate > 8%.
14:18	Triage	On-call engineer confirmed HTTP 502 responses across candidate and recruiter endpoints.
14:22	Identification	Isolated two triggers: Gunicorn worker exhaustion and failing payment webhooks.
14:26	Mitigation	Issued emergency service restarts across Gunicorn and Celery daemons.
14:32	Rollback	Executed Git rollback of the latest payment webhook release (v1.4.2 -> v1.4.1).
14:38	Data Recovery	Executed administrative replay scripts to reprocess unacknowledged payment transactions.
14:45	Verification	Synthetic health checks and mock interview WebSockets returned green status.
14:48	Resolution	Incident declared resolved; metrics returned to normal thresholds.

Deliverable 2: Root Cause Analysis (RCA)

2.1 The "5 Whys" Investigation Model

1. Why did the frontend return 502 Bad Gateway errors?

Nginx was unable to communicate with Gunicorn because the Unix domain socket backlog was completely saturated.

2. Why was the Gunicorn socket saturated?

All 3 Gunicorn WSGI workers were blocked and killed by Gunicorn's 30-second internal timeout monitor.

3. Why were the Gunicorn workers blocked?

Multiple candidates submitted complex 8-page PDF resumes simultaneously through a synchronous parsing endpoint instead of the asynchronous Celery pipeline.

4. Why did payment webhooks fail at the same time?

A recent deployment expected a strict customer_email key in the incoming webhook body. When the payment gateway sent a payload missing that optional parameter, the unhandled KeyError crashed the thread inside an uncommitted database transaction.

5. Why did voice interview connections drop?

Under high CPU load from the runaway PDF parsing tasks on the application server, the local STT bridge exceeded its socket timeout buffer, causing the media streamer to terminate candidate connections.

2.2 Root Cause Summary

Primary Cause (Infrastructure): Heavy I/O and CPU-bound resume parsing tasks ran synchronously inside HTTP worker threads instead of being routed to isolated Celery worker processes.

Secondary Cause (Application Code): Brittle JSON payload parsing in the billing webhook listener lacked schema validation and defensive .get() fallbacks, triggering unhandled exceptions that failed database commits.

2.3 Permanent Fix Documentation & Code Patches

Patch A: Defensive Webhook Consumer (core/views.py)

Replace brittle dictionary key indexing with defensive extraction and error isolation:

```
# Before (Vulnerable)
customer_email = payload['customer_email']
sub = UserSubscription.objects.get(user__email=customer_email)

# After (Hardened & Resilient)
customer_email = payload.get('customer_email') or payload.get('data', {}).get('object',
{}).get('email')
if not customer_email:
    logger.error("Webhook payload missing identifier: %s", payload)
    # Return 200 to acknowledge delivery to the gateway, but flag for manual review
    return Response({"status": "ignored", "reason": "missing_identity"}, status=status.HTTP_200_OK)
```

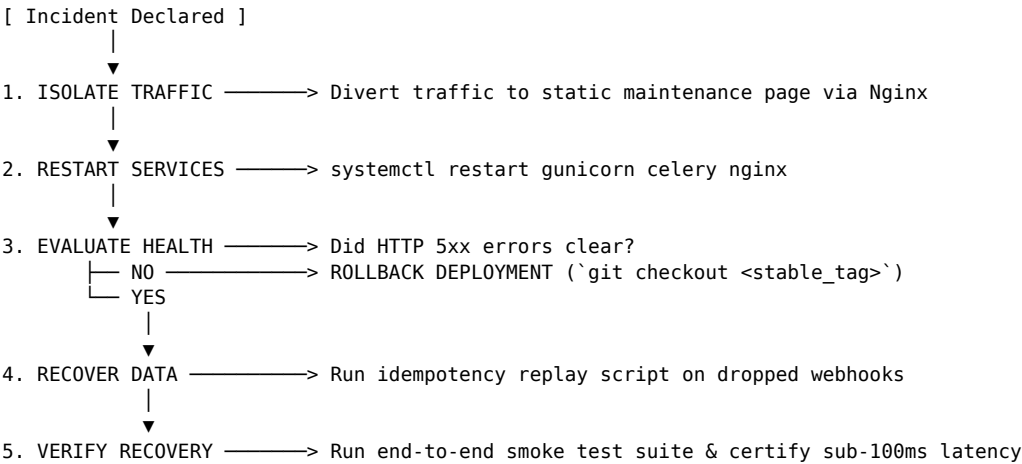
Patch B: Asynchronous Enforcement

Route all document parsing requests to the background task queue:

```
# Enforce asynchronous processing via Celery
task = parse_candidate_resume_task.delay(s3_file_key=s3_key, user_id=request.user.id)
return Response({
    "status": "queued",
    "task_id": task.id,
    "message": "Resume uploaded successfully. Processing in background."
}, status=status.HTTP_202_ACCEPTED)
```

Deliverable 3: Production Recovery & Continuity Plan

PRODUCTION SYSTEM RECOVERY WORKFLOW



Preventive Measures & Action Items

Priority	Category	Action Item	Owner	Target Due Date
P1	Architecture	Enforce strict async delegation for all PDF parsing via Celery; add hard request size caps (10MB) in Nginx.	DevOps Team	Immediate
P1	Billing	Implement JSON schema validation and dead-letter queues (DLQ) for all Stripe and Razorpay webhook events.	Backend Lead	Immediate
P2	Monitoring	Configure Datadog/Prometheus alerts for Gunicorn worker queue depth (>5 requests) and Celery task latency (>5s).	SRE Team	3 Days

Priority	Category	Action Item	Owner	Target Due Date
P2	AI Pipeline	Deploy a dedicated worker pool for speech-to-text processing to isolate audio streaming from standard API traffic.	AI Team	5 Days